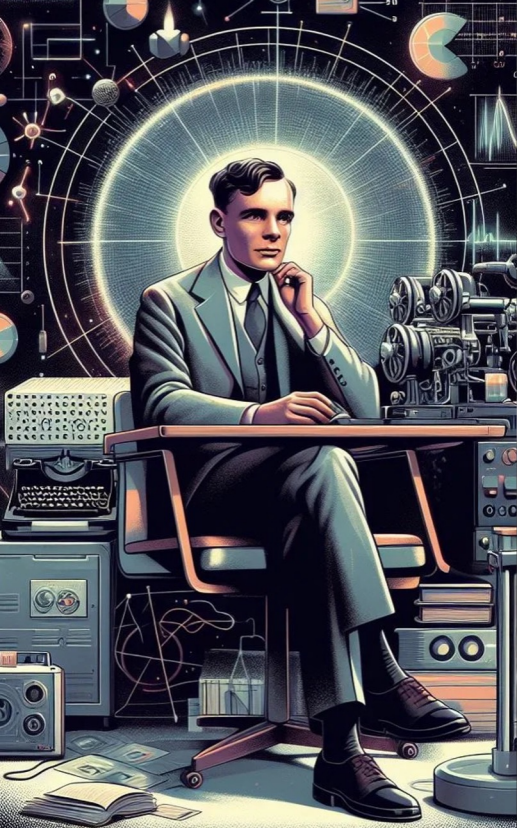




Lecture 1: Introduction to the Course, Classical Cryptography

CS409: Introduction to Cryptography
Instructor: Sruthi Sekar

Nice

iPhone 15



Try!



What Cryptography Will Teach You

- What primitives/tools to use for which task?
 - Encryption (Confidentiality)
 - Digital Signatures (Authenticity)
 -
- How to define and prove security of these primitives?
- Is this security unconditional or under certain assumptions?
- Trade-offs between efficiency and security for real-world needs.
- Essential backbone of secure systems

A man, likely Jimmy Kimmel, is smiling and holding a microphone on a city street. He is wearing sunglasses and a black t-shirt. His right hand is raised to his forehead. The background shows a busy street with pedestrians, cars, and buildings. The text "JIMMY Kimmel LIVE!" is overlaid on the bottom left.

JIMMY
Kimmel
LIVE!

What Cryptography Can't Protect You From

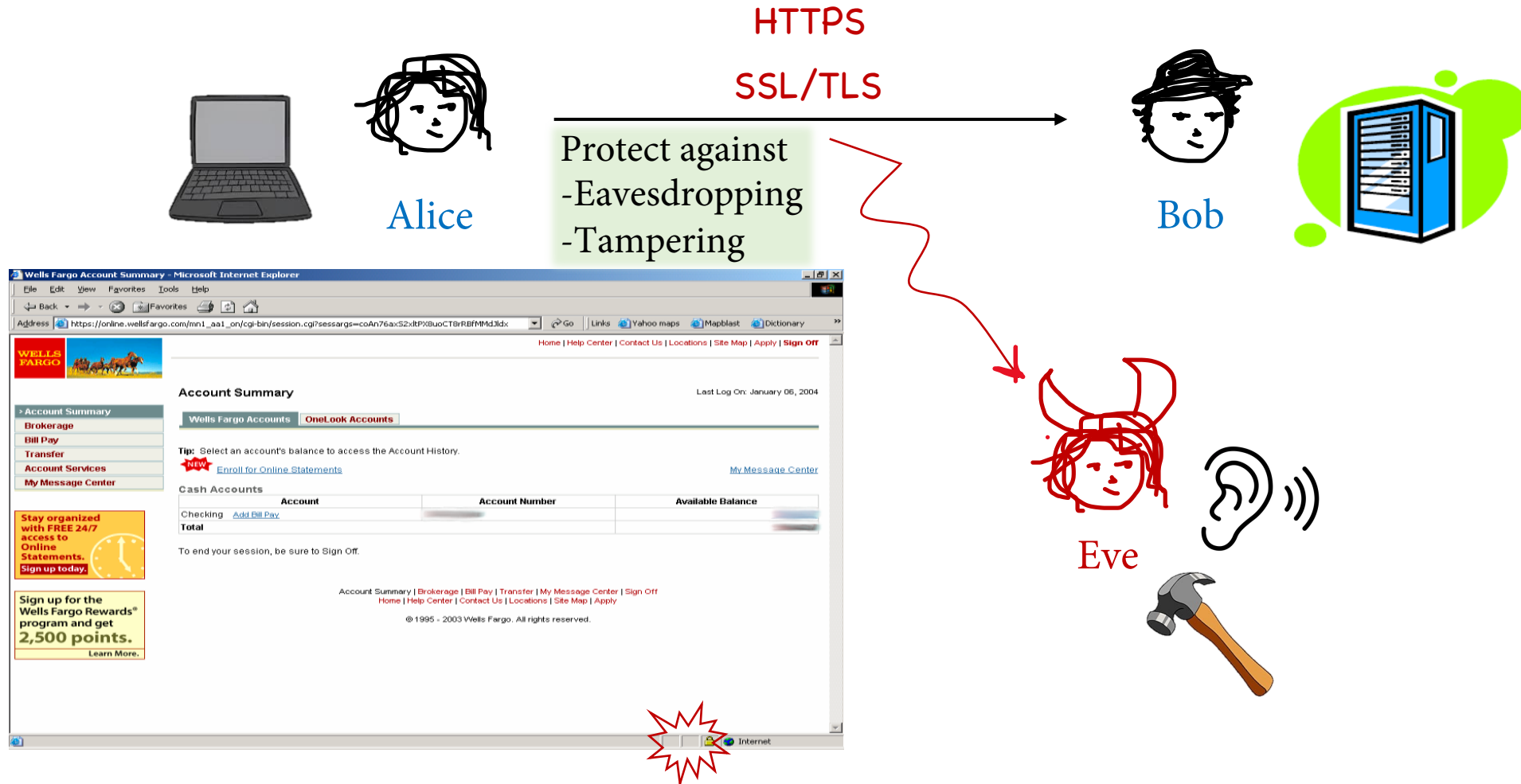
- **When you use it:** It does not solve all security problems
 - Software bugs
 - Social engineering attacks (phishing attacks, trusting wrong person/interface, etc.)
- **When you implement it:** Reliable only if implemented and used correctly.
 - E.g., WEP contains many mistakes.
- **When you build it:** Rely on well-studied standard primitives instead of inventing your own designs and assumptions.
 - [Microsoft's Dual EC DRBG](#) (used in Windows): was a random number generator.
 - Custom authentication schemes in IoT devices: smart locks, cameras, etc. encrypt using shifting ciphers (generally rely on “security by obscurity”).

You Use Cryptography Everywhere!

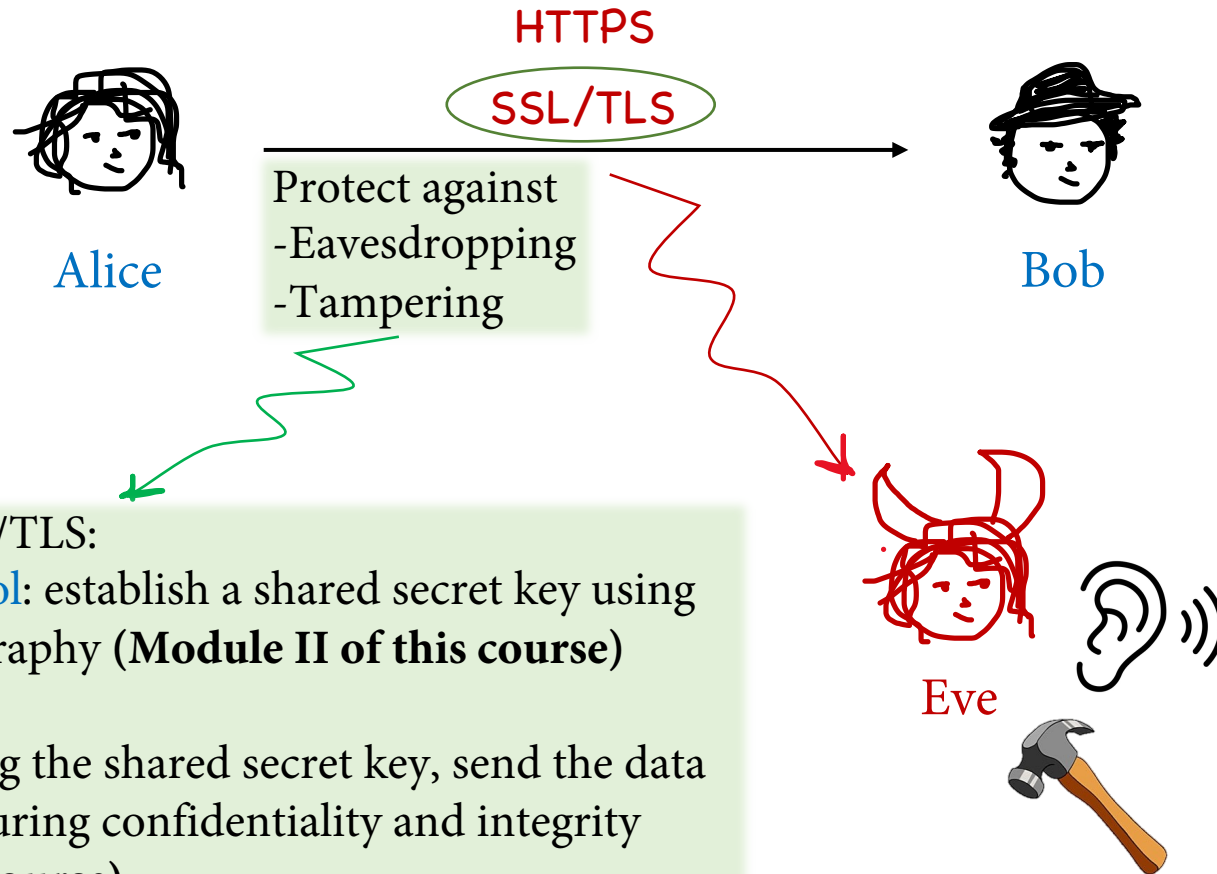
- **Secure communication:**
 - web traffic: HTTPS
 - wireless traffic: 802.11i WPA2 (and WEP), GSM, Bluetooth
 - WhatsApp, Signal, Proton mail
- **File and Disk Encryption:** EFS, TrueCrypt, LUKS
- **Content Protection** (e.g. DVD, Blu-rays): CSS, AAC
- **User Authentication:** password hashing, Multi-factor authentication, biometric authentication, FIDO/WebAuthn, etc.
- **Finance:** UPI, Cryptocurrencies, Credit/Debit Cards (EMV)
- Many many more....



Secure Communication



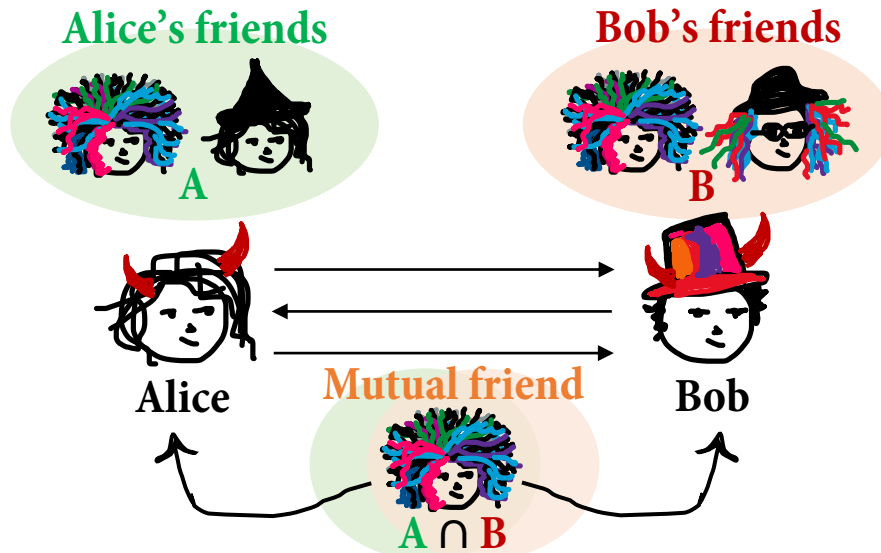
Secure Communication



Example of an Advanced Tool (Module III)

Private Set Intersection (PSI)

Alice and Bob want to jointly compute intersection of their private sets.



Security Need: Alice (resp. Bob) should not learn any information about Bob's (resp. Alice's) set, except the intersection.

The Apple PSI System

Abhishek Bhowmick
Apple Inc.

Dan Boneh
Stanford University

Steve Myers
Apple Inc.



Kunal Talwar
Apple Inc.

Karl Tarbe
Apple Inc.

July 29, 2021



Signal

The Difficulty Of Private Contact Discovery

moxie0 on 03 Jan 2014



Building a social network is not easy. Social networks have value proportional to their size, so participants aren't motivated to join new social networks which aren't already large. It's a paradox where if people haven't already joined, people aren't motivated to join.



Password Checkup extension

Offered by: [google.com](https://www.google.com)

★★★★★ 295 | [Productivity](#) | 900,000+ users

By Google

WIRED

BUSINESS CULTURE GEAR IDEAS SCIENCE SECURITY TRANSPORTATION SIGN IN SUBSCRIBE

LILLY MAY NEWMAN SECURITY 06.19.2019 09:00 AM

Google Turns to Retro Cryptography to Keep Data Sets Private

Google's Private Join and Compute will let companies compare notes without divulging sensitive information.

Plan For This Course

MODULE I

Secure Communication
in **Shared-Key Setting**

Symmetric Key Encryption

Message Authentication

Cryptographic Hash Functions

MODULE II

Secure Communication
in **Public-Key Setting**

Public Key Encryption

Digital Signatures

MODULE III

Some Advanced Tools

Zero Knowledge Proofs

Fully Homomorphic Encryption

Private Set Intersection

Course Objectives and Recommendations

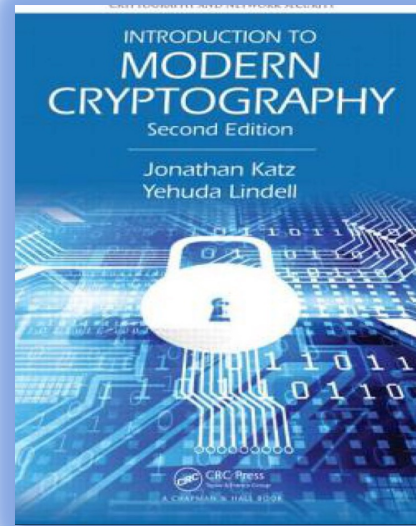
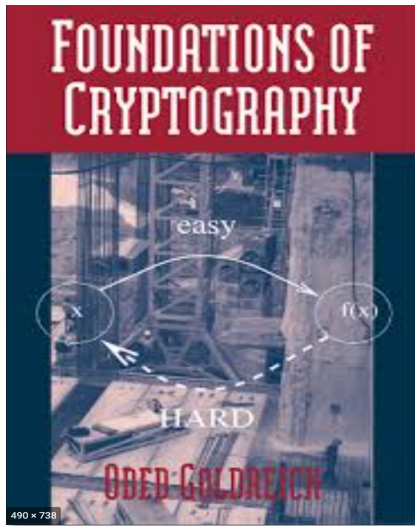
- **Course Objective:**

- Learn how cryptographic primitives work.
- Understand the security notions formally and write proofs of security.

- **Recommendations:**

- Be active in class, take a few notes, and refer to slides. Primary references will be class slides with extra references mentioned on the next slide and on the course webpage.
- All questions are important, so ask as many questions as possible in class. I will try to either answer them right away or get back to you with an answer.
- Solve the assignments (even though they are not graded). Lab exercises will help you apply what you learn! Try to make the most of this course!

Course Homepage and References



[A Graduate Course in Applied Cryptography](#)

Dan Boneh
Victor Shoup

COURSE HOMEPAGE (for references/slides/announcements/grading policy): [link](#)

Course Logistics

Grading Policy

Attend and be interactive to get 5%

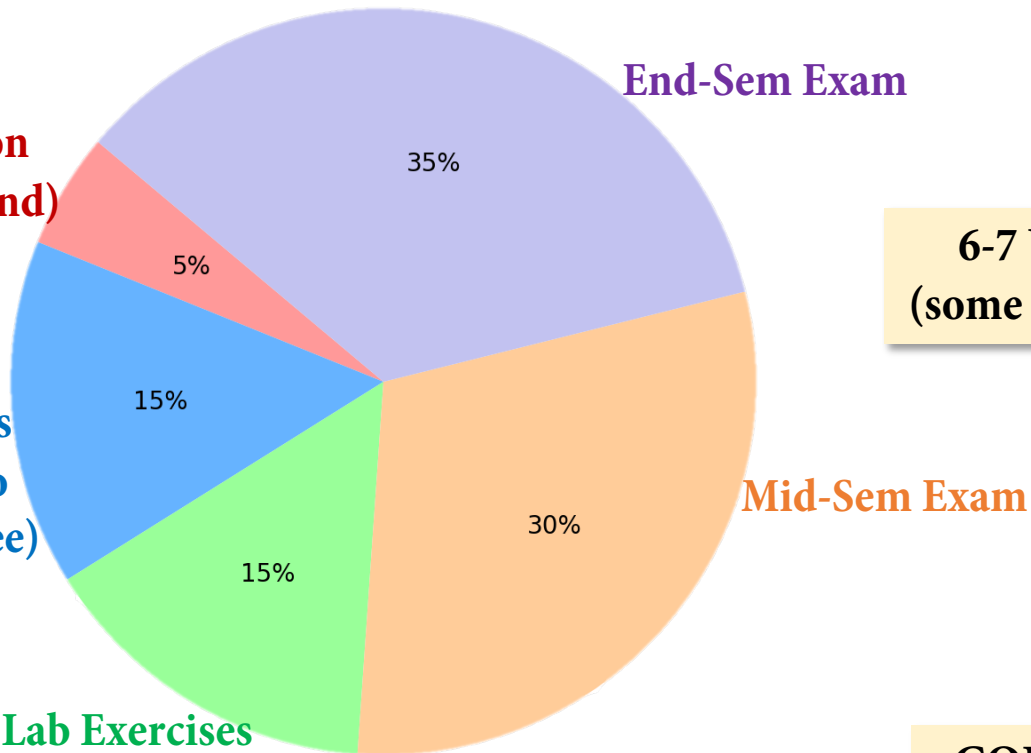
Class Participation
(Ask, Answer, Attend)

Quiz 1: 3rd Sept, CC103
(in Quiz slot: 8:30-9:30 AM)

3 Quizzes
(best two
out of three)

Lab Exercise 1: 8th Aug (tentatively)
(Prep material I: Out today)

4 Lab Exercises



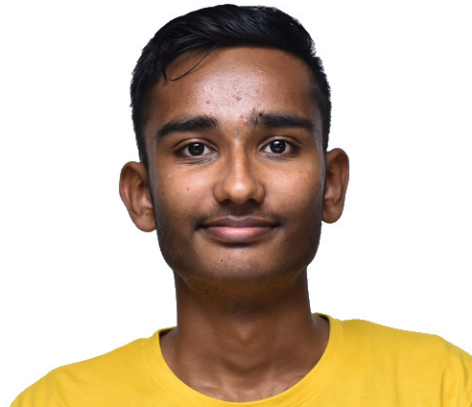
6-7 Ungraded Assignments
(some questions in Quiz/Exams)

COURSE HOMEPAGE: [link](#)

MOODLE: additional discussions, lab exercise submissions, grades

Contact Hours and TAs

- **Meet me after class**, or **email me** to set up an appointment (sruthi@cse.iitb.ac.in).
- **Teaching Assistants**: will be your help through the course (regular doubt sessions)!



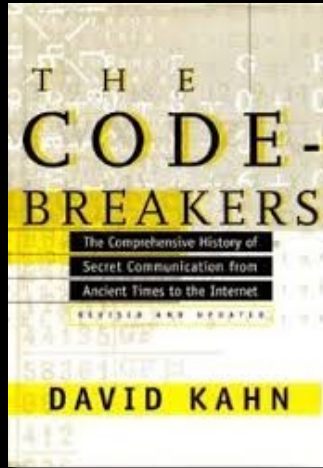
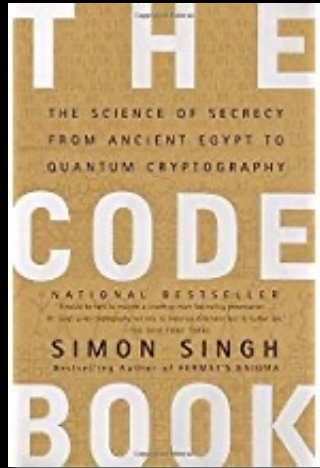
Lakshya
22b0995@iitb.ac.in



Ravi
24m2131@iitb.ac.in

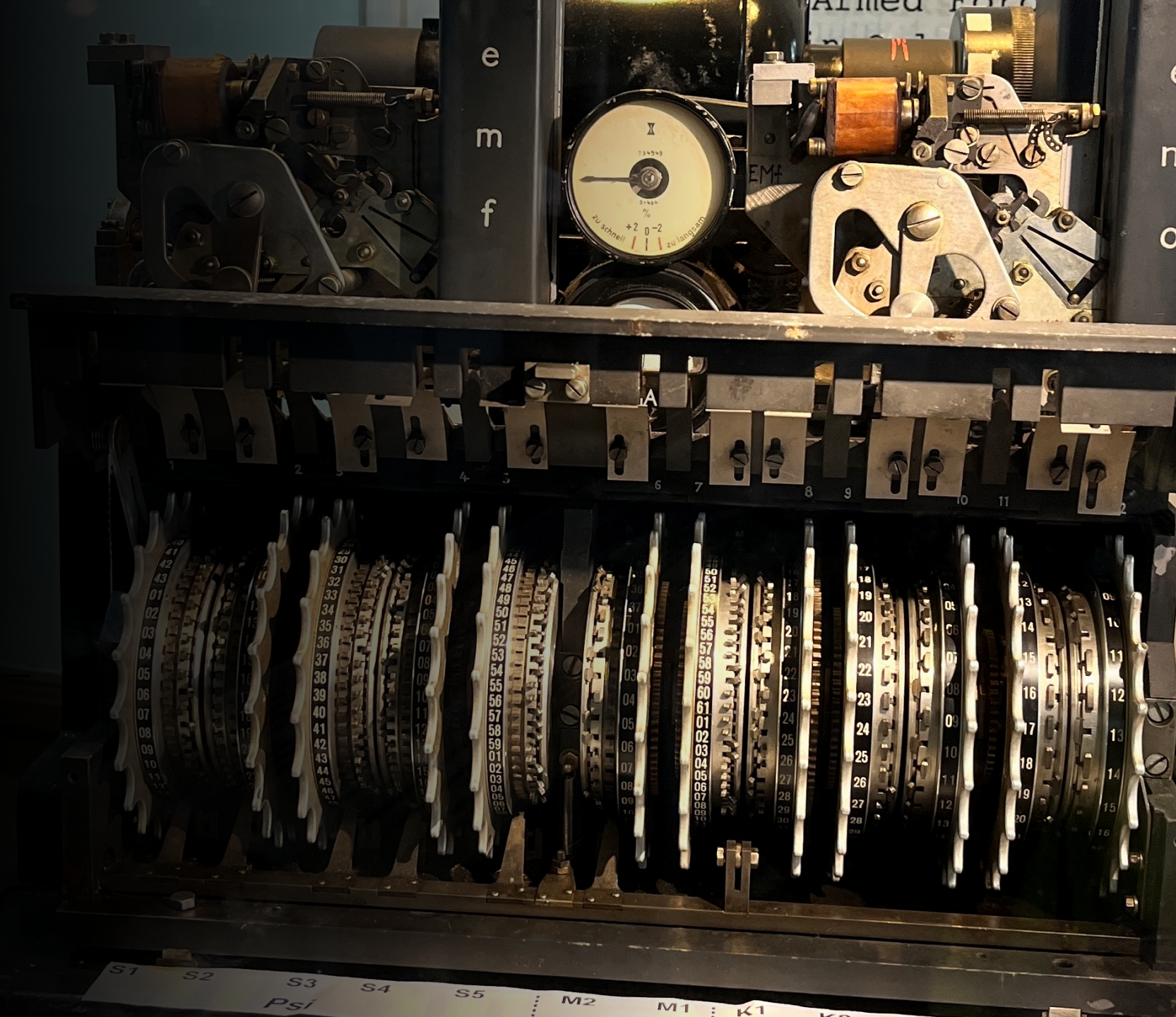
- **Temporary Class Rep**: the goal is to find a TA discussion time for assignment solving. (form a group, add TAs to it, and finalize a slot to meet them weekly).

Volunteer(s)?



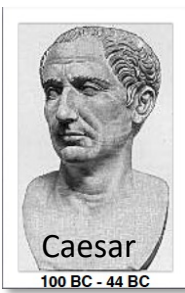
SOME HISTORY

Welcome to
BLETCHLEY
Home of the Codebreakers





1900 BC
Hieroglyphics



Caesar
100 BC - 44 BC



Al-Kindi
801-873

Classical vs Modern



Claude Shannon



Alan Turing



Shamir, Rivest & Adleman



Merkle, Hellman & Diffie

1900 BC

110 BC

WWII

1980s

Classical Crypto

Modern Crypto



What?:

- ✓ Code Design
- ✓ Code Breaking



How?:

- ✓ Steganography, Caesar Cipher, Enigma, Substitution cipher, etc.



Who used it?:

- ✓ Kings, Queens, Military Organizations, etc.

What?:

- ✓ **Pillar 1:** Formal definition of security capturing the requirements.
- ✓ **Pillar 2:** Precise, well-studied assumptions to rely on.
- ✓ **Pillar 3:** Rigorous mathematical proof of security.



How?:

- ✓ Secure Encryptions, Authentication, Secure Computation, etc.

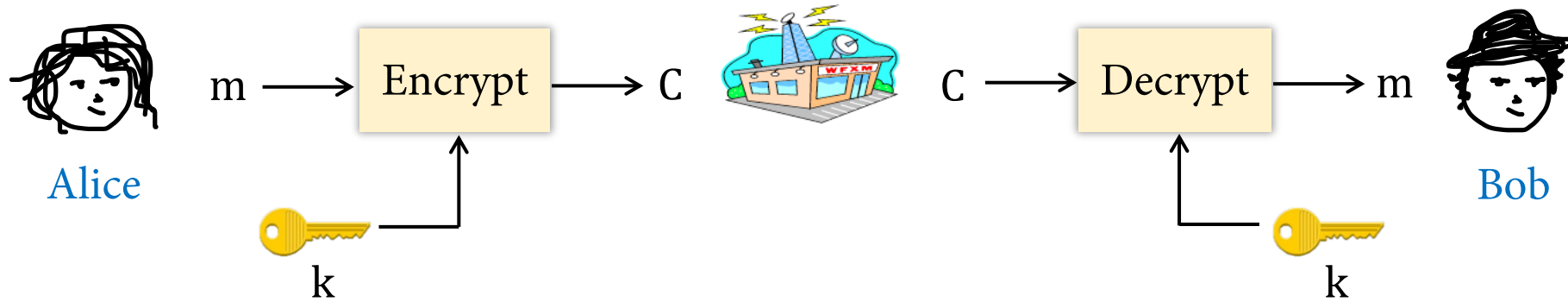


Who uses it?:

- ✓ Everyone! (Net-banking, e-purchase, software updates, email)



Symmetric Encryption



m - Plaintext message comes from some message space (set) $\mathcal{M}$

C - Ciphertext (hidden message) lies in some ciphertext space $\mathcal{C}$

k - Shared common key (shared in advance “somehow”) from key space $\mathcal{K}$

symmetry



Caesar's Cipher [~58BC]



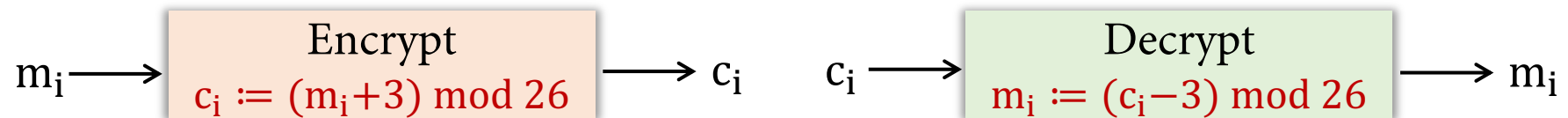
Encryption Algorithm:

Shift By 3

msg	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	
cipher	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	

No key

$\mathcal{M} = \mathcal{C} = \{0, 1, 2, \dots, 25\}$ (interpret the alphabets as numbers)



- No secret key! Trivial to break!
- Encryption/Decryption must remain private here!



Lesson

Don't hide Enc/Dec

Kerckhoffs' Principle

The cipher method must not be required to be a secret, and it must be able to fall into the hands of the enemy without any inconvenience.

Security should only rely on the secrecy of the key!

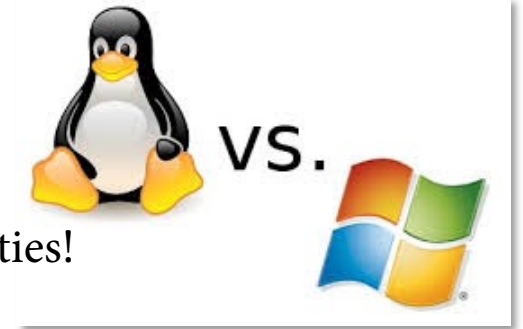


Auguste Kerckhoffs
19th-century
Dutch cryptographer

Kerckhoffs is right because..



1. Maintaining the privacy of a short key (≈ 100 bits) is easier than maintaining the privacy of a large algorithm (1000 times larger).
2. Easy to replace the key than a whole program if exposed!
3. It is infeasible to imagine a secret pair of algorithms for every pair of communicating parties!



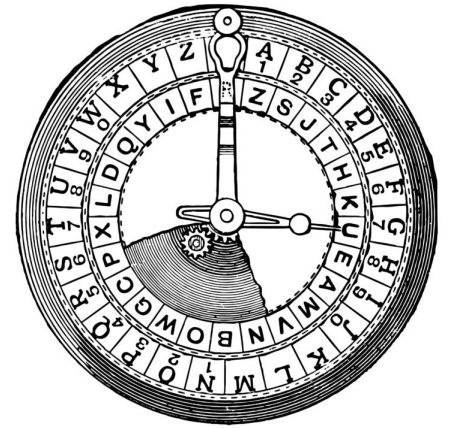
More reasons to have an open cryptographic design:

1. Open designs undergo public scrutiny and, hence, are likely to be stronger!
2. Security flaws (if they exist) can be revealed by “ethical” hackers.
3. Public design enables establishment of standards.



**Dangerous to use proprietary
Encryption schemes!**

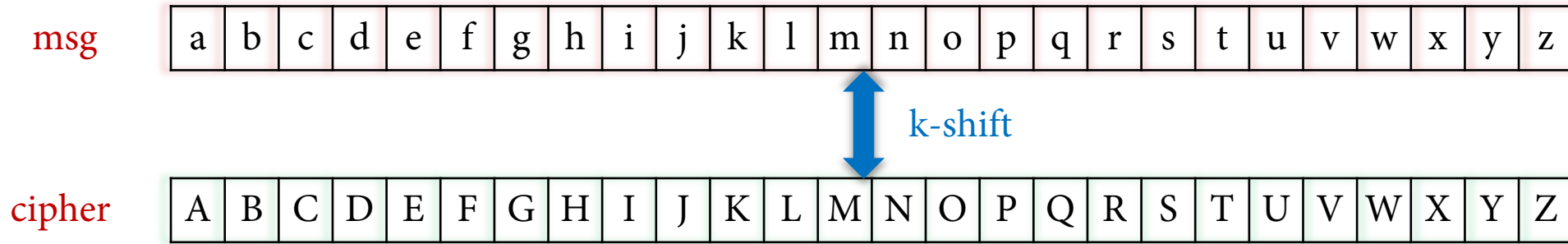
Shift Cipher [~58BC]



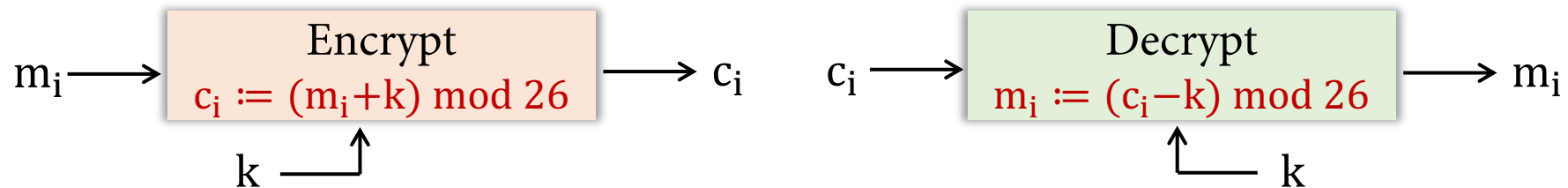
Key space: $\mathcal{K} = \{0, 1, \dots, 25\}$. Key k chosen at random from $\mathcal{K}$.

Encryption Algorithm:

Shift By k



$\mathcal{K} = \mathcal{M} = \mathcal{C} = \{0, 1, 2, \dots, 25\}$ (interpret the alphabets as numbers)



BROKEN



Brute-force Attack- try all 26 keys!

Lesson

Sufficient key-space principle

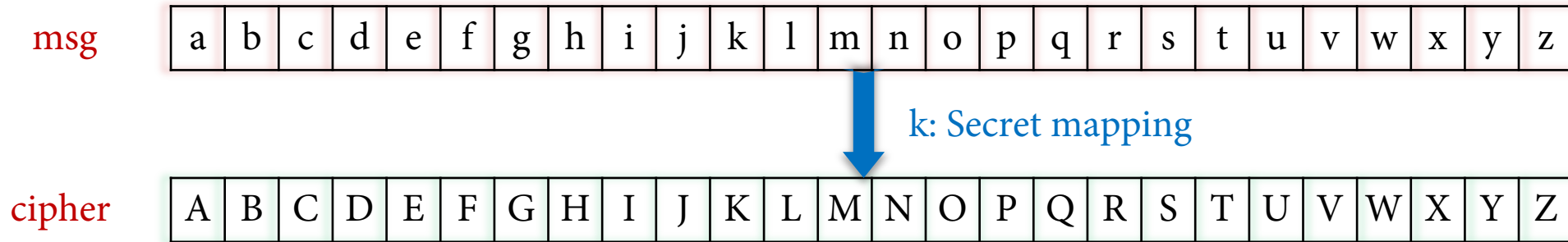
Mono-alphabetic Substitution Cipher [1300s]



Key space: $\mathcal{K}$ = all permutations on $\{0,1, \dots 25\}$. Key k chosen at random from $\mathcal{K}$.

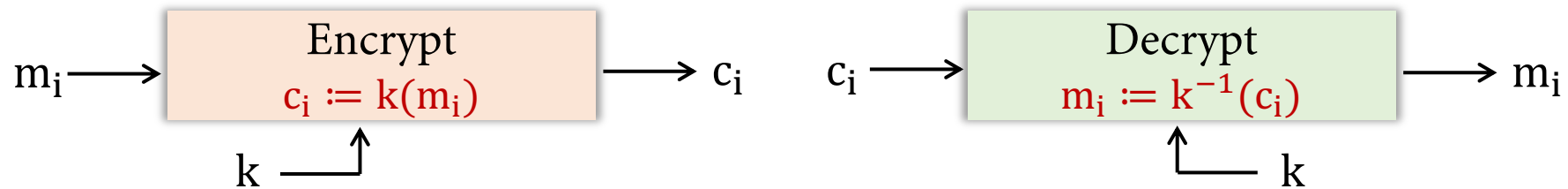
Encryption Algorithm:

Shift using the map k



$\mathcal{M} = \mathcal{C} = \{0,1,2, \dots, 25\}$ (interpret the alphabets as numbers)

$\mathcal{K} = \{k: \{0,1,2, \dots, 25\} \rightarrow \{0,1,2, \dots, 25\}: k \text{ is a permutation}\}$



$|\mathcal{K}| = 26! \approx 2^{88}$, so no brute-force attack possible!

Mono-alphabetic Substitution Cipher [1300s]



BROKEN



- Easy break: Frequency/Statistical Analysis (exploiting statistical patterns of the English language)

EXAMPLE

Ciphertext-only attack

UKBYBIPOUZBCUFEEBORUKBYBHOBBERFESPVKBWFOFERNBCVBZPRUBOFERNBCVBPCYYFVUFO
FEIKNWFRFIKJNUPWRFIPOUNVNIPUBRNCUKBEFWWFDNCHXCYBOHOPYXPUBNCUBOYNRVNIWN
CPOJIOFHOPZRVFZIXUBORJRUBZRBCHNCBBONCHRJZSFWNVVRJRUBZRPCYZPUKBZPUNVPWPCYVF
ZIXUPUNFCPWRVNBCVBRPYYNUNFCPWWJUKBYBIPOUZBCUIPOUNVNIPUBRNCHOPYXPUBNCUB
OYNRVNIWNCPOJIOFHOPZRNCRVNBCUNENVVVFZIXUNCHPCYVFZIXUPUNFCPWZPUKBZPUNVR



HW: Break this cipher!

B	36	→ E
N	34	
U	33	→ T
P	32	→ A
C	26	

NC	11	→ IN
PU	10	→ AT
UB	10	
UN	9	

digrams

UKB	6	→ THE
RVN	6	
FZI	4	

trigrams

Lesson

Don't fix mapping from $\mathcal{M}$ to $\mathcal{C}$!



Blaise de
Vigenère

Vigenère (Poly-alphabetic Shift) Cipher [1553]

- 💡 “Smooth out” the frequency distribution of characters in the ciphertext, making it harder to perform statistical analysis.
- 💡 Key now maps each instance of plain-text character to different ciphertext characters.

Key: k is a random word of length t .

Encryption Algorithm: say $k = \text{CRYPTO}$ ($t = 5$)

msg

w	h	a	t	a	n	i	c	e	d	a	y	t	o	d	a	y
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

+ mod 26

key

C	R	Y	P	T	O	C	R	Y	P	T	O	C	R	Y	P	T
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

cipher

Z	Z	Z	J	U	C	L	U	D	T	U	N	W	G	C	Q	S
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

HW: Break this.
1. Try known t .
2. Freq analysis++

BROKEN

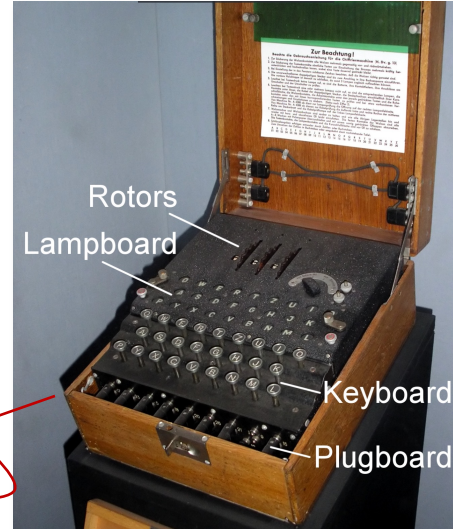
Was harder to break, systematic attack took hundreds of years to device!

Rotor Machines

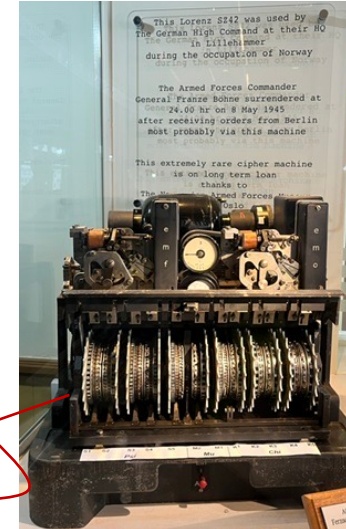


Hebern Machine
(single rotor)

Some form of
statistical analysis
used to break them!



Enigma
(3-5 rotors)

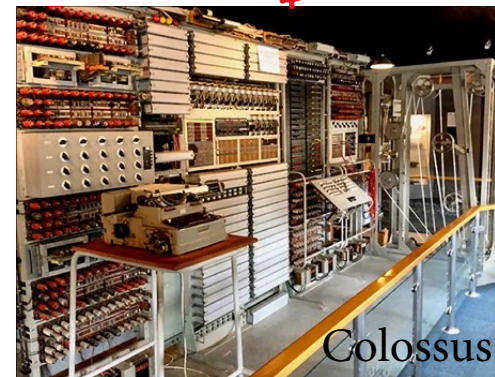


Lorenz SZ42
(12 rotors)

Breaking these
ciphers led to the
first computers



The Turing Bombe



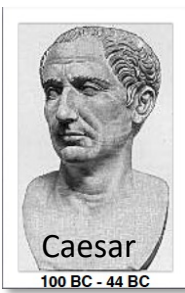
Colossus

Lesson

No Adhoc designs!



1900 BC
Hieroglyphics



Caesar
100 BC - 44 BC



Al-Kindi
801-873

Classical vs Modern

1900 BC

110 BC

WWII

1980s



Claude Shannon



Alan Turing



Shamir, Rivest & Adleman



Merkle, Hellman & Diffie

Classical Crypto



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How?:

- ✓ Secure Encryptions, Authentication, Secure Computation, etc.

Who uses it?:

- ✓ Everyone! (Net-banking, e-purchase, software updates, email)

Modern Crypto



Modern Cryptography

1. Identify the threat model (adversary) and formalize what it means to defeat the adversary. This becomes your security model.
2. Most modern cryptographic constructions cannot be proven unconditionally and are based on certain assumptions (unproven/conjectured statements that have been tested for a long time).
3. Give a concrete mathematical proof of security in the well-defined security model identified in step 1, either unconditionally or based on some assumption.

Prep For Next Class: Please revise probability basics before next class
(References: A.3 from Katz-Lindell, [lecture slides](#))

Prep For Lab Exercises to Come: will post some prep-material through Lab Exercise 0 **today** (ungraded). Please familiarize yourself with it and contact the TAs if you need help!



THANK YOU